

ALLIANCE AGCM, AB, Canada

WMO: 710060

Lat: 52.3154N

Lon: 111.7787W

Elev: 2418

StdP: 13.46

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.7	-17.3	-28.6	1.2	-22.4	-23.2	1.7	-17.3	29.1	28.9	26.0	23.9	6.6	150	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	84.4	62.5	80.8	61.2	77.7	60.4	66.0	77.7	64.0	76.0	62.4	74.1	10.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	90.9	70.9	59.9	84.3	68.7	58.1	78.8	66.7	32.1	77.5	30.5	76.0	29.3	74.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	23.1	20.2	DB	-31.2	90.6	5.3	4.3	-35.0	93.7	-38.1	96.2	-41.1	98.6	-45.0	101.7	(4)
(5)			WB	-31.4	70.5	5.3	2.4	-35.2	72.3	-38.3	73.7	-41.3	75.0	-45.1	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	14.6	12.9	24.5	38.0	51.2	58.2	62.9	61.1	52.4	39.9	24.8	12.8		
	DBStd	21.45	15.55	14.95	14.14	9.49	7.88	5.74	4.74	5.64	8.37	8.99	12.93	14.06		
	HDD50	5700	1097	1040	789	370	83	8	0	2	70	329	758	1154		
	HDD65	9959	1562	1460	1254	810	431	212	98	143	385	778	1207	1619		
	CDD50	1290	0	0	0	11	119	255	399	347	141	17	1	0		
	CDD65	74	0	0	0	0	2	10	32	24	6	0	0	0		
	CDH74	1643	0	0	0	6	126	196	576	522	212	5	0	0		
	CDH80	393	0	0	0	0	16	27	156	135	59	0	0	0		
	WSAvg	9.2	10.2	9.5	9.7	10.6	10.0	8.9	8.0	7.5	8.2	9.5	9.2	9.3		
	PrecAvg	16.1	0.8	0.5	0.7	1.2	1.7	2.9	2.8	2.2	1.3	0.7	0.7	0.7		
(15) Precipitation	PrecMax	21.5	2.4	0.9	1.7	2.8	4.1	4.3	5.9	4.9	3.3	1.9	2.4	1.2		
	PrecMin	10.3	0.2	0.1	0.0	0.2	0.4	0.7	0.7	0.2	0.0	0.2	0.1	0.1		
	PrecStd	3.3	0.5	0.3	0.4	0.6	0.9	1.0	1.2	0.9	0.9	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	46.5	58.3	71.7	82.5	83.4	89.3	88.2	86.3	73.4	58.4	45.7		
		MCWB	38.2	38.7	45.8	49.6	56.7	62.2	65.6	62.5	60.1	54.0	45.8	38.5		
	2%	DB	41.6	40.4	51.7	66.0	77.5	79.2	84.4	83.5	79.8	65.3	50.5	40.4		
		MCWB	36.2	34.9	41.5	48.1	54.8	60.3	64.5	62.6	58.0	50.3	41.3	34.7		
	5%	DB	38.7	36.8	45.7	59.8	73.3	75.7	79.9	79.9	74.8	59.9	44.4	36.0		
		MCWB	34.6	32.8	38.6	44.4	53.2	59.3	63.4	61.6	56.5	47.5	37.2	31.6		
	10%	DB	35.7	33.6	41.1	54.6	68.7	72.0	76.5	76.1	69.3	54.8	40.8	31.5		
		MCWB	32.0	30.7	35.6	41.9	51.3	57.2	62.4	60.8	54.3	44.5	35.3	28.5		
	0.4%	WB	38.7	39.1	46.8	51.9	59.0	65.4	70.3	68.2	62.3	56.1	46.9	39.3		
		MCDB	44.2	46.3	57.3	69.8	77.0	77.9	81.3	78.3	80.2	71.2	58.6	44.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.6	35.2	42.0	48.2	56.4	62.7	66.8	65.4	59.4	51.7	42.0	34.9		
		MCDB	41.0	39.4	49.9	64.6	73.0	74.1	78.4	76.7	76.0	60.5	49.6	39.8		
	5%	WB	34.7	33.2	38.7	45.3	54.5	60.7	64.9	63.3	57.2	48.4	38.4	31.8		
		MCDB	38.7	36.7	45.1	57.7	70.5	72.2	76.8	75.7	71.4	58.0	43.6	35.7		
	10%	WB	32.2	30.7	35.9	42.8	52.4	58.8	63.3	61.6	55.3	45.5	35.7	28.7		
		MCDB	35.6	33.7	40.6	53.3	66.4	69.4	74.5	73.6	68.9	54.3	39.6	31.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	17.9	18.2	21.9	25.0	22.4	23.8	24.5	25.1	21.4	17.4	16.5		
		MCDBR	17.5	19.8	23.1	31.8	32.4	28.1	31.3	32.8	34.7	29.8	22.0	19.2		
		MCWBR	13.4	15.4	14.4	16.3	14.6	12.3	14.4	14.7	15.9	16.6	14.2	14.8		
		MCDBR	18.0	19.6	22.5	29.2	29.3	24.5	27.1	27.8	31.3	27.8	20.7	19.0		
	5% WB	MCWBR	14.4	15.4	14.6	15.7	13.7	12.1	13.9	14.1	15.1	15.9	14.0	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.301	0.291	0.333	0.357	0.365	0.359	0.374	0.318	0.285	0.260	0.250		
	taud		2.277	2.245	2.431	2.357	2.336	2.350	2.404	2.341	2.458	2.532	2.460	2.324		
	Ebn at Noon		233	256	289	287	284	281	281	270	276	266	242	222		
	Edh at Noon		20	28	28	35	37	37	35	35	28	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		351	685	1139	1509	1834	1908	2019	1643	1158	694	356	262		
	RadStd		31	54	85	140	121	127	99	127	119	89	35	24		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page